Problem Set 1: Education and Political Economy

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Problem 1: Education

A school counselor was curious about the average IQ of students in her school and took a random sample of 25 students' IQ scores:

```r
# Load libraries (none needed here)
rm(list=ls())
detachAllPackages <- function() {
  basic.packages <- c(
  "package:stats", "package:graphics", "package:grDevices",
  "package:utils", "package:datasets", "package:methods", "package:base"
  )
  package.list <- search()[ifelse(unlist(gregexpr("package:", search()))==1,
     TRUE, FALSE)]
  package.list <- setdiff(package.list, basic.packages)
  if (length(package.list) > 0) {
    for (package in package.list) detach(package, character.only=TRUE)
  }
}
detachAllPackages()
y <- c(
105, 69, 86, 100, 82, 111, 104, 110, 87, 108,
87, 90, 94, 113, 112, 98, 80, 97, 95, 111,
114, 89, 95, 126, 98
)
# 1. 90% Confidence Interval
n <- length(y)
sample_mean <- mean(y)
sample_sd <- sd(y)
alpha <- 0.10    # 90% CI
t_crit <- qt(1 - alpha/2, df = n - 1)
se <- sample_sd / sqrt(n)
ci_lower <- sample_mean - t_crit * se
ci_upper <- sample_mean + t_crit * se
cat("Sample mean:", sample_mean, "\n")
cat("Sample SD:", sample_sd, "\n")
cat("90% CI for population mean: [", ci_lower, ",", ci_upper, "]\n")
# 2. One-sample t-test (test if mean > 100)
t_test_result <- t.test(y, mu = 100, alternative = "greater", conf.level =
  0.95)
print(t_test_result)
```

Listing 1: R code for Problem 1

Part 1: 90% Confidence Interval for the Mean

The sample mean and standard deviation, calculated from the data, are:

- $\bar{y} = 98.44$
- $s = 13.09$

The 90% confidence interval is given by:

$$\left[\bar{y} - t_{\alpha/2,n-1}\frac{s}{\sqrt{n}},\ \bar{y} + t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}\right] = [93.96,\ 102.92]$$

Therefore, we are 90% confident that the true mean IQ score of students in this school is between 93.96 and 102.92.

Part 2: Hypothesis Test

We test:

$$H_0:\ \mu = 100,$$
$$H_A:\ \mu > 100$$

R output:

```
One Sample t-test
data:  y
t = -0.59574, df = 24, p-value = 0.7215
alternative hypothesis: true mean is greater than 100
95 percent confidence interval:
93.95993      Inf
sample estimates:
mean of x
98.44
```

Since p-value $= 0.7215 \gg 0.05$, we **fail to reject** the null hypothesis. There is insufficient evidence to conclude that the average IQ at this school exceeds the national average of 100.

Summary (Q1)

- 90% confidence interval: [93.96, 102.92]
- p-value $= 0.72$ ($t = -0.60$). The school's mean IQ is **not significantly higher** than the national average 100.

Problem 2: Political Economy

Researchers are curious about what affects the amount communities spend on homelessness. The dataset has 50 US states (see code for variable descriptions).

1. Preparation and Data Import

```r
rm(list=ls())
detachAllPackages <- function() {
  basic.packages <- c("package:stats", "package:graphics", "package:grDevices", "package:utils",
  "package:datasets", "package:methods", "package:base")
  package.list <- search()[ifelse(unlist(gregexpr("package:", search()))==1, TRUE, FALSE)]
  package.list <- setdiff(package.list, basic.packages)
  if (length(package.list)>0) for (package in package.list) detach(package, character.only=TRUE)
}
detachAllPackages()
# Install/load ggplot2
pkgTest <- function(pkg){
  new.pkg <- pkg[!(pkg %in% installed.packages()[,"Package"])]
  if (length(new.pkg)){ install.packages(new.pkg, dependencies = TRUE)}
  sapply(pkg, require, character.only = TRUE)
}
pkgTest(c("ggplot2"))
# Data import (demo - fill in with actual data as needed)
  expenditure <- read.table("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2025/main/datasets/expenditure.txt", header=T)
str(expenditure); summary(expenditure); head(expenditure)
```

Listing 2: R code for data import

Variables:

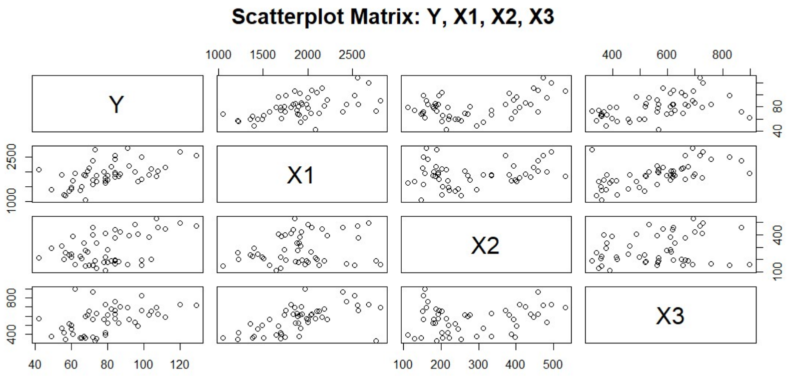
- Y: Per capita expenditure on shelters/housing assistance
- X_1: Per capita income
- X_2: Residents per 100,000 "financially insecure"
- X_3: Per 1,000 living in urban areas
- Region: 1=Northeast, 2=North Central, 3=South, 4=West

2. Exploratory Data Analysis

```r
pairs(expenditure[,c("Y","X1","X2","X3")],
main="Scatterplot Matrix: Y, X1, X2, X3")
cor_mat <- cor(expenditure[,c("Y","X1","X2","X3")])
print(round(cor_mat, 2))
```

Listing 3: Scatterplot matrix and correlation

Figure 1: Scatterplot Matrix



Correlation output:

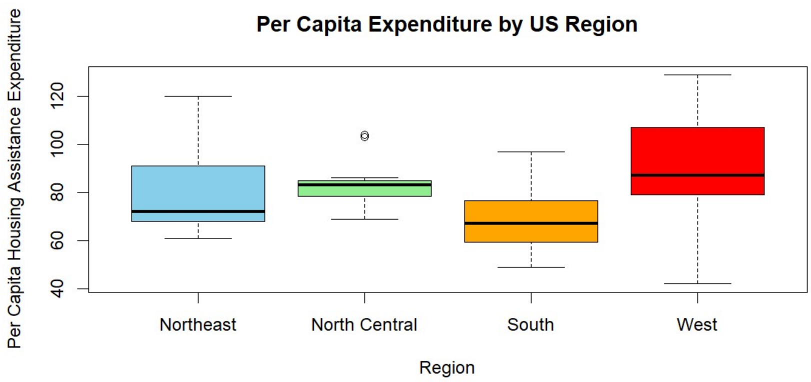
```
Y    X1   X2   X3
Y  1.00 0.53 0.45 0.46
X1 0.53 1.00 0.21 0.60
X2 0.45 0.21 1.00 0.22
X3 0.46 0.60 0.22 1.00
```

3. Housing Expenditure by Region

```r
region.lab <- c("Northeast","North Central","South","West")
boxplot(Y ~ Region, data = expenditure, names = region.lab,
ylab = "Per Capita Housing Assistance Expenditure",
xlab = "Region",
main = "Per Capita Expenditure by US Region",
col = c("skyblue","lightgreen","orange","red"))
region.means <- tapply(expenditure$Y, expenditure$Region, mean)
print(region.means)
```

Listing 4: Boxplot for Y by Region

Figure 2: Boxplot of Y by Region



Region means:

1	2	3	4
79.44	83.92	69.19	88.31

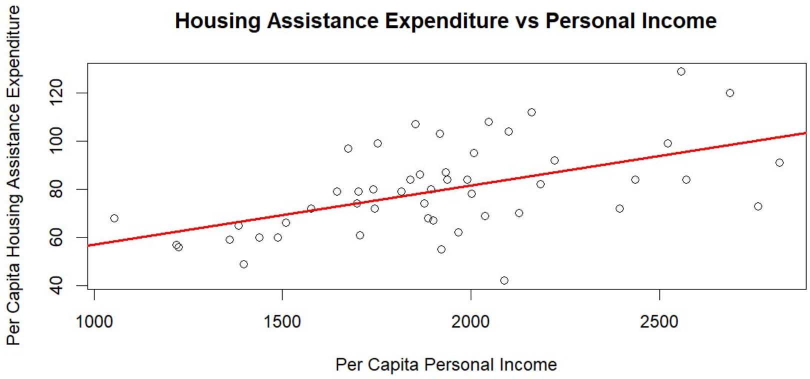
Interpretation: West region (Region 4) has the highest mean per capita expenditure.

4. Expenditure vs Personal Income

```
plot(expenditure$X1, expenditure$Y,
xlab = "Per Capita Personal Income",
ylab = "Per Capita Housing Assistance Expenditure",
main = "Housing Assistance Expenditure vs Personal Income")
abline(lm(Y ~ X1, data = expenditure), col="red", lwd=2)
```

Listing 5: Scatterplot and regression line

Figure 3: Expenditure vs Income



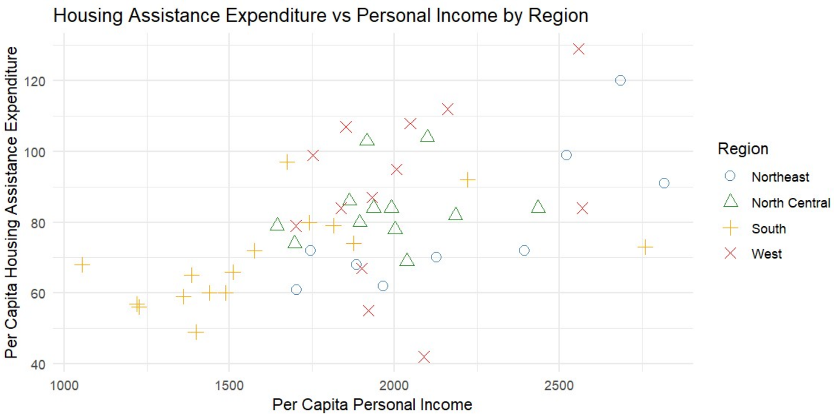
Interpretation: Higher per capita income tends to associate with higher housing assistance expenditure.

5. Expenditure vs Income by Region

```
library(ggplot2)
ggplot(expenditure, aes(x = X1, y = Y, color = factor(Region), shape = factor(
    Region))) +
geom_point(size = 3) +
labs(
title = "Housing Assistance Expenditure vs Personal Income by Region",
x = "Per Capita Personal Income",
y = "Per Capita Housing Assistance Expenditure",
color = "Region", shape = "Region"
) +
scale_color_manual(values = c("steelblue","forestgreen","orange","red"),
labels = region.lab) +
scale_shape_manual(values = 1:4, labels = region.lab) +
theme_minimal()
```

Listing 6: Scatterplot by region with ggplot2

Figure 4: Expenditure vs Income (by Region)



Interpretation: The West (red) tends to spend more, even at similar income levels, indicating regional effects.

Summary (Q2)

- West region has the highest average per capita expenditure.
- Expenditure rises with income, but region is an independent factor.
- For similar incomes, Western states still spend more than others.